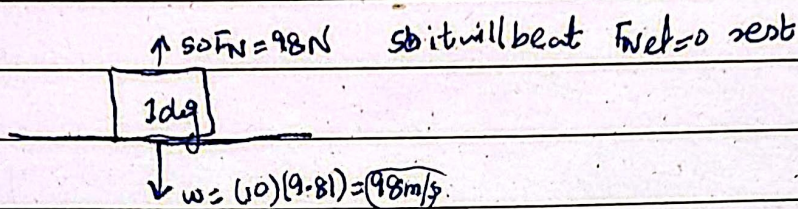


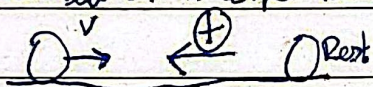
Newton's first law:-

It states an object at rest remains at rest, "An object in motion remains in motion" until an external force is applied.



Newton's second law:-

If someone throw a ball on a carpet it will slow down after less time.



But on smooth surface like ice ball will still move

← Air resistance

long time

due to less friction.

if object at rest, $F_{\text{net}} = 0$

If object is moving with constant velocity → $F_{\text{net}} = 0$, $a = 0$

but if $F_{\text{net}} \neq 0$ so $a_{\text{net}} \neq 0$

$$a = \frac{v_f - v_i}{t} = \frac{0}{t} = 0\text{ m/s}^2$$

$m \uparrow, a \downarrow$
 $m \downarrow, a \uparrow$

$$F_{\text{net}} = ma$$

$$m \uparrow \quad F_{\text{net}} \uparrow \quad (\text{more})$$

$$a \uparrow \quad F_{\text{net}} \uparrow \quad (\text{more})$$

$$F_{\text{net}} = m \left(\frac{v - u}{\Delta t} \right)$$

$$F_{\text{net}} = m \frac{\Delta v}{\Delta t}$$

$$P = mv$$

at rest, $P = 0$

at not rest, $P \neq 0$

$$F_{\text{net}} = \frac{\Delta P}{\Delta t}$$

$$\text{Impulse} = \boxed{F \Delta t}$$

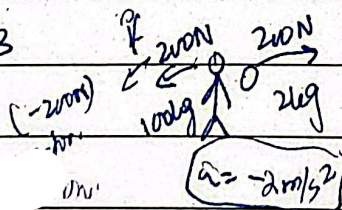
(m/s)

momentum

Newton 3rd law:-

$$F_A = -F_B$$

$m \downarrow, a \uparrow$



$$a_{\text{of ball}} = F = ma$$

$$200 = m(a) \rightarrow a = +1\text{ m/s}^2$$

TYPES OF Forces

Noncontact

Gravitational

Electric

Magnetic

Contact

Tension

Friction

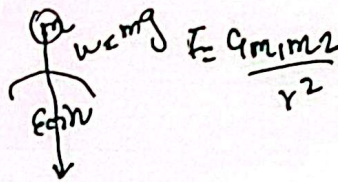
Normal reaction force

Spring force.

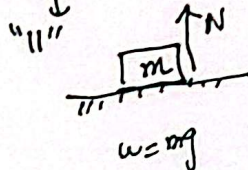
Gravitational force ($1kgf = 9.8N$)

(weight)

acts towards the centre of earth



Normal Reaction (N)



"like if earth is attracting object towards centre but if object don't move so a force will act in upwards direction"
"|| to surface at point of contact"

Tension (T) string/Rod



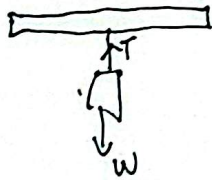
string change
T → change

Freebody diagram (a). A box at rest on a road.

$$F_N = w$$



(b). A block attached to a rope hangs from ceiling.



$$T = mg$$

(c). A ball is upward at constant velocity



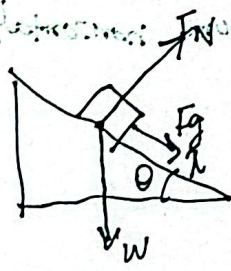
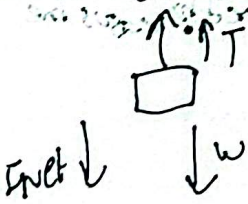
$$a = 0$$
$$\Sigma F_y = T - w$$
$$ma = T - w$$
$$0 = T - w$$
$$T = w$$

(d). A box upward with constant acceleration.

$$\Sigma F_y = T - w$$
$$ma = T - mg$$
$$m(a + g) = T$$

so Tension now bigger.

(e). if descend with a downward acceleration



$$F_N = mg - T_y$$

$$a_x = T_x - f_k$$

$$F_N = mg \cos \theta$$
$$\Sigma F_x = F_g$$
$$F_g = mg \sin \theta$$
$$a = g \sin \theta$$

$$F_{net} = 0, a = 0$$
$$v = 20, t = ?, v = 8$$
$$v = u + at$$
$$500 = v + (10)t$$
$$300 - 20 = 280$$
$$100 = 20a$$
$$u = 0, a = 6$$
$$t = 12$$
$$F - 35 = 4$$

$$50 \times 5$$
$$150N$$
$$0 = 100 + 250 - F_N$$

$$F_N = F + 100 - 200$$
$$F_N = w - T_y$$
$$F_N = mg - T \sin \theta$$

Elevator Q's

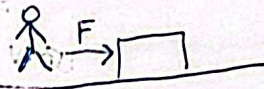
If moving with constant velocity $\rightarrow a=0$

If moving with constant velocity upward/down $\rightarrow a=0$

If moving with some accel upward $\rightarrow \Delta F = m[g + ay]$

downward $\rightarrow \Delta F = m[g - ay]$

CH 16



If a box is at stationary to move

If you need to put harder force to move it

so it is a force that prevents the box from sliding when you exert



and if object starts to move then it is

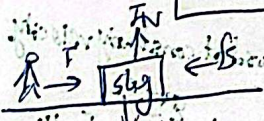
f_k (kinetic friction)

$f_s \leq \mu_s F_N$ (represent a range of motion)

$$f_k = \mu_k F_N$$

$$\mu_s > \mu_k$$

Q. $\mu_s = 0.40$
 $\mu_k = 0.2$



$$W = 5 \times 9.8 = 49 \text{ N}$$

$$f_s \leq (49) \times (0.4) \leq 19.6 \text{ N}$$

$$f_k = 49 \times 0.2 = 9.8 \text{ N}$$

F_A	f_s	f_k	F_{net}
0	0	0	0
10	10	0	0
15	15	0	0
19.6	19.6	0	0
20	0	9.8	10.2

always will $f_s < 19.6$

The applied force will be equal to static friction

otherwise 0

"In the second case applied is greater than kinetic friction but it will be 0 as object will still remain static"

After that it will be equal to its value $\rightarrow f_k$

horizontal force that is required to cause the block to

Q. A 35 kg rest on horizontal. what is minimum static coefficient of static friction is 0.35?

Ans:-

$$f_s = F(u)$$

$$F = f_s$$

$$F = 49 \text{ N}$$

$\rightarrow F$ using

$$F = 51.45 \text{ N}$$

$$F > 51.45 \text{ N}$$



(b). what is acceleration of system if a person pushes the box with a force of 90N. if coefficient of kinetic friction is 0.20?

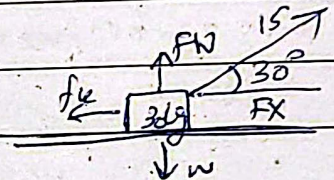
$$F_k = 0.2(90) \rightarrow$$

$$90 - (15 \times 9.81 \times 0.2) = 15a \rightarrow a = 4.04 \text{ m/s}^2$$

Q3). A force of 150N pulls the 30kg of box to right. coefficient of kinetic friction is 0.25. what is horizontal acceleration of box?

$$F_N \neq mg$$

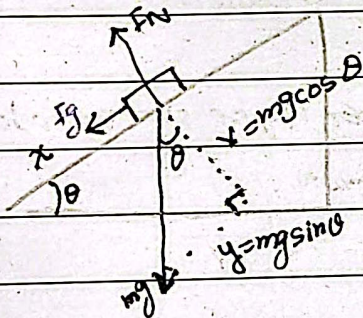
$$150 \cos 30 - (30 \times 9.81 \times 0.25) = 30a$$



$$F_N + 15 \sin 30 = mg$$

$$F_N = 219 \text{ N}$$

Inclined plane:



$$1. \text{ Normal force} = mg \cos \theta$$

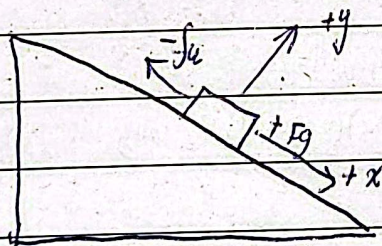
$$\text{So } F_g = mg \sin \theta$$

$$F_k = mg \sin \theta$$

$$a = g \sin \theta$$

"if a block rest on a frictionless inclined plane height acceleration we obtain"

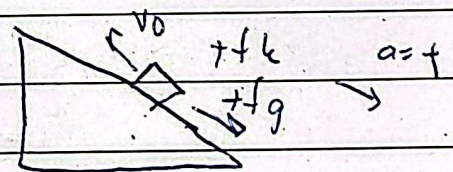
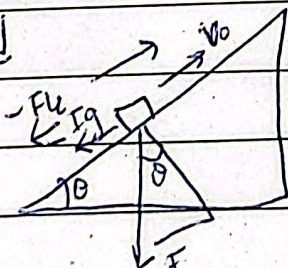
So F_k opposite



$$E_{fx} = \text{Forward} - \text{Back}$$

$$m a_{\text{max}} = mg \sin \theta - \mu_k F_N$$

if going up



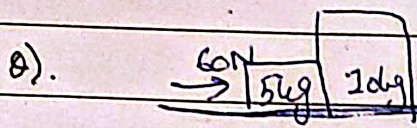
$$-F_k - F_g = ma$$

$$-\mu_k F_N - mg \sin \theta = ma$$

$$a = -$$

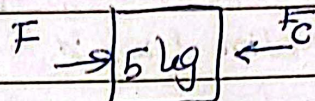
GENIUS

Contact forces :-



$$a = \frac{EF}{m} \rightarrow \frac{60}{15} = (4) \text{ m/s}^2$$

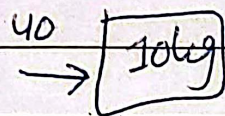
"if a force is applied on 5kg it will also apply force on 10kg where reaction force will be applied on 5kg"



$$EFx = F - F_c$$

$$ma = F - F_c$$

$$F = ma + F_c \rightarrow F_c = F - ma = 60 - (5)(4)$$

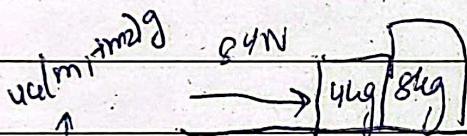


$$F_c = (10)(4) = 40 \text{ N} \rightarrow \text{so contact force is } 40 \text{ N}$$

$$F_c = 40$$

if no kinetic friction.

Q). calculate contact force exerted by each block on other figure shown below. coefficient of kinetic friction b/w a block and horizontal surface is 0.20.

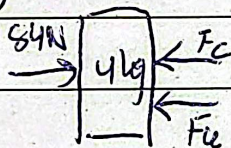


$$a = 5.4 \rightarrow (7) \times$$

Note: wrong because didn't see μ_k .

$$a = \frac{F - F_k}{m_1 + m_2} = \frac{84 - (0.2 \times 12 \times 9.8)}{12}$$

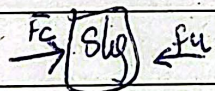
$$a = 5.04$$



$$ma = 84 - (F_c + F_k)$$

$$(4)(5.04) = 84 - (F_c + 0.2 \times 4 \times 9.8)$$

$$F_c = 56 \text{ N}$$

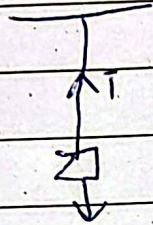


$$F_c = (8 \times 0.2 \times 9.8) = (8 \times 5.04) = 56 \text{ N}$$

Tension forces

Q). A rope is used to lift a 50 kg box with an upward acceleration of 2.3 m/s^2

(a) Tension rope (b). what would be Tension if rope if box descended slowly with downward $a = 0.75 \text{ m/s}^2$?

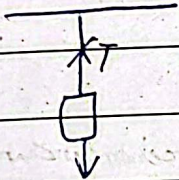
Ans:-

$$T - (50 \times 9.81) = (50 \times 2.3)$$

$$T = 605 \text{ N}$$

if asked $T > W$

(b).

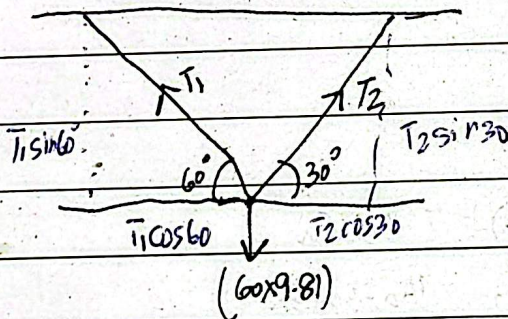


$$(50 \times 9.81) - T = (50 \times 0.75)$$

$$T = 50 \times (9.81 - 0.75) \rightarrow 455.5 \text{ N}$$

 $T < W$

Q). Tension in ropes:

Answer:-

$$\Sigma F_x = T_2 \cos 30^\circ - T_1 \cos 60^\circ$$

$$0 = T_2 x - T_1 x$$

$$T_1 x = T_2 x$$

$$T_1 \cos 60 = T_2 \cos 30$$

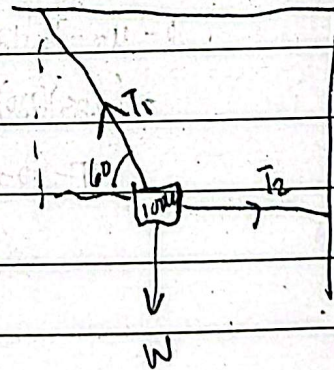
$$T_1 = 1.73 T_2$$

$$\Sigma F_y = T_1 \sin 60 + T_2 \sin 30 - W$$

$$0 = T_1 y + T_2 y - W$$

Then solve simultaneously

Q).

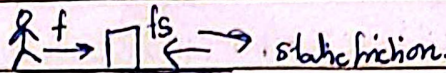
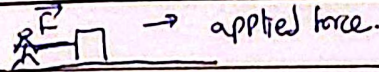
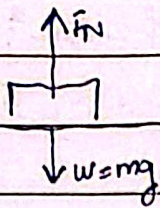


$$F_y = T_1 \sin 60 - W$$

$$0 = T_2 - T_1 \cos 60$$

GENIUS

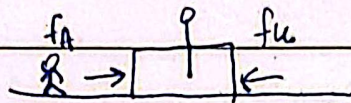
Different types of forces



if Normal force increased so static friction increases so chances of little drifting of box.

if Normal force is decreased so friction decreases.

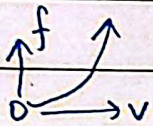
if start to move



$$f_A = F_A > F_{smax}$$

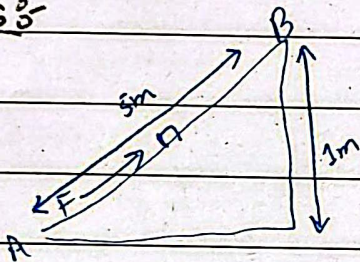
$$f_k = \mu_k F_N$$

if force, velocity same direction $\Rightarrow$ so it increases.



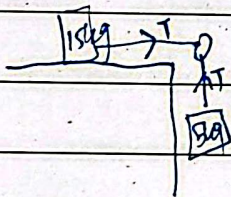
if Perpendicular (Then No effect on magnitude of force it's direction can change).

Simple mechanics 3-



This 3, 5 tells that $5 \times 1 = 5$ times less force on inclined

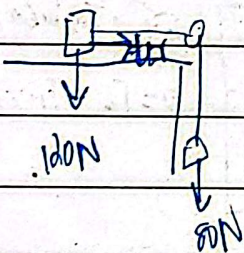
Pulleys



$$T = 15a \quad -T = 15a \quad T - 50 = 5a$$

$$T = 15a, \quad -15a - 50 = 5a$$

$$-50 = 20a \Rightarrow a = -2.5 \text{ m/s}^2$$



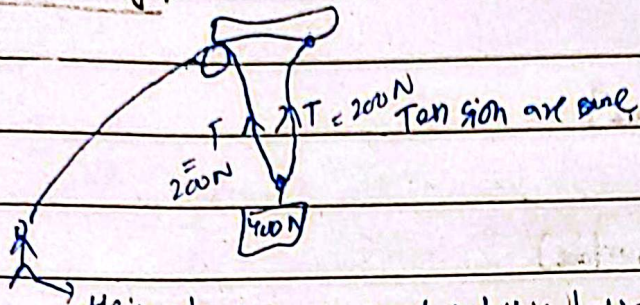
$$0 - 40 - T = 120a$$

$$(0.25)(120) + T = -120a$$

$$T - 80 = 8a$$

$$a = -4.5, \quad T = 36N$$

Simple machine lifting 950



$$MA = 2$$

$$\frac{400}{200} = 2$$

$$F_{out} = 400N \uparrow$$

He is applying 200N object but it is 400N up. as

$$F_{in} = 200N \downarrow$$

1W = same

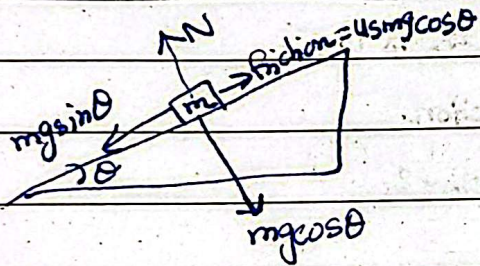
and distance get multiplied by MA.

Lever :-

$$F_1 d_1 = F_2 d_2$$

Gravitational acceleration :-

Angle of repose on inclined plane :-



Max θ at which block stays at rest

θ is at which the box is about to slide

$$\theta_R = \tan^{-1}(\mu_s)$$

in this when we increase the θ , $\sin \theta$ increases $\cos \theta$ decreases.

Given $\theta < \theta_R$ [No sliding] [Rest] so static friction = applied force

Given $\theta = \theta_R$ [Just about to slide] so limiting friction $\rightarrow F_L = 45N$

Given $\theta > \theta_R \rightarrow$ Then L.C.

if μ is then $a = (g \sin \theta) - (\mu g \cos \theta)$

θ . $2g$ $\mu_s = 0.4$

$\mu_k = 0.3$

$11.88, 6.28$ (4.7)

$11.8 - 4.7 = 7.1$

(3.5)

GENIUS

$a = 1.735$